POKHARA UNIVERSITY

Level: Bachelor | Semester – Fall | Year : 2013
Programme: B.E | Full Marks: 100
Course: Microprocessors | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Describe the evolution of Microprocessor. The processing speed of Harvard architecture is higher than Von Neumann machine, Explain Why? 8
   b) Draw the functional block diagram of 8085 microprocessor and explain the functions of each block in detail. 7
2. a) Draw and explain the timing diagram for the instruction LDA 9000H 8
   b) Write an ALP for 8085 to count the integers available in an array starting from memory location 2500H to 2510H for exactly divisible by two and save the result in the register B. 7
3. a) Explain the addressing modes of 8086 with suitable examples. 8
   b) Write an 8086 ALP which will input the student's registration number from the keyboard. If the registration number is valid it will display the students name else it will display 'Not enrolled' 7
4. a) The six data bytes are stored from memory location 3000H to 3005H. Write an 8086 ALP to transfer block of data to new location 5000H to 5005H. 7

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   b) Draw an address decoding circuit for 8085 to interface 2KB RAM and 4KB ROM. 8
5. a) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standard? 7

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b) What is Interrupt priority of 8085 Microprocessor? Write an assembly language program to check to see if RST 6.5 n pending, assuming the microprocessor completing RST 7.5 interrupt request. If it is pending, enable RST 6.5 without affecting any other interrupts; otherwise, return to the main program. 8

6. a) Explain how the 20 bit physical address is calculated in 8086 Microprocessor? Compare 8086 Microprocessor with 8085 Microprocessor 8

b) WAP in 8085 to transfer ten bytes data from 5050H to 5060H only if data is between 30H and 70H else store 00H in the next table. 7

7. Write short notes on **any two:** 2×5

a) Macro assembler (8086)

b) Flags of 8086

c) RISC vs. CISC

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